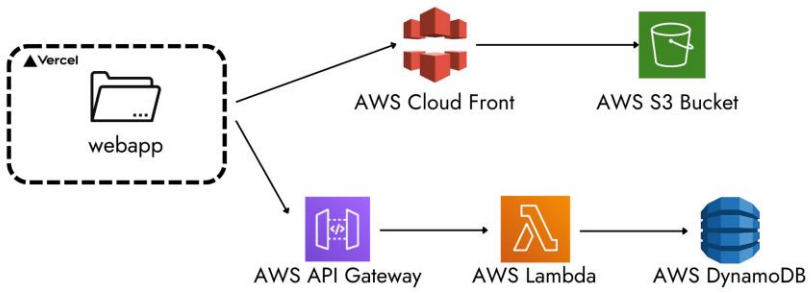
Background

The education landscape is rapidly evolving, with an increasing demand for flexible and affordable learning solutions. The project, *EduPath* aims to bridge the gap by developing a robust ed-tech platform that offers free courses and guided career paths. By providing learners with access to comprehensive educational resources and personalized career guidance, we seek to revolutionize the way people acquire knowledge and skill sets.



To achieve this, the project will leverage on numerous **AWS services**, including AWS Lambda, DynamoDB, API-gateway, CloudFront, S3. These five play crucial roles in the development and implementation of *EduPath* as the platform can benefit from serverless compute capabilities, simplified development workflows, scalable backend services, and efficient data storage and retrieval. DynamoDB will be used as the primary database storage, supported by the Lambda function and API-Gateway to call different functions from platform. On the other hand, Vercel will be *EduPath's* frontend web application manager, providing scalable infrastructure for hosting and deploying the application.

Overall, the project aims to bridge the knowledge gap and create opportunities for individuals to access quality learning resources. At the same time, enhancing the usage of cloud technologies for both frontend and backend deployment.

Implementation Details

App Description

EduPath is a simple ed-tech streaming platform and career guidance service built using cloud technologies. The report will be focusing on the technologies used and how it has been integrated to the cloud platform.

Frontend Technology

- **ReactJS** [https://react.dev/]

React.js is utilized as the frontend technology to create a modular and reusable user interface. By breaking down the UI into components, React enables efficient development and maintenance. Through its virtual DOM and declarative syntax, React optimizes rendering performance and simplifies UI updates based on

application state. With its robust state management capabilities, React ensures effective handling of user interactions and dynamic content rendering. The extensive React ecosystem offers a wide range of libraries and tools, empowering the team to enhance the platform's functionality and user experience.

The frontend application of this EdTech platform is responsible for delivering an intuitive and user-friendly interface to learners, instructors, and administrators. It consists of various components, including user interfaces, interactive elements, and visual elements like layouts, menus, and forms.

Key features of the frontend application include:

1. Course Navigation: Users can easily browse and navigate through available courses, accessing detailed course information, syllabi, and learning resources. They can enroll in courses, track their progress, and manage their learning journey.
2. Content Presentation: The frontend presents educational content in a structured and visually appealing manner, such as video lectures, interactive quizzes, reading materials, and multimedia resources. It allows users to access and consume content seamlessly, supporting different learning styles.
3. Personalization: The frontend application provides personalized experiences by tailoring content and recommendations based on user preferences, learning history, and performance. It may offer recommendations for additional courses, adaptive learning paths, or personalized feedback to enhance the learning process.

Backend Technologies

- **AWS DynamoDB** [https://aws.amazon.com/dynamodb/]

Amazon DynamoDB is utilized as the frontend database technology to store and manage data. DynamoDB's scalable and fully managed NoSQL database service allows for seamless storage and retrieval of data. The flexible schema and automatic scaling capabilities enables the platform to handle especially high workloads and accommodate the growing user base. I utilize this technology for the purpose of storing *courses*, *careers*, and *awan-users.*

- **AWS Lambda** [https://aws.amazon.com/lambda/]

Amazon Lambda is leveraged as the frontend serverless technology to handle various backend functions as response to API calls. Lambda functions handle incoming user requests together with the communication with AWS DynamoDB without the need for server provisioning, allowing the platform to provide a responsive and efficient frontend experience for its users.

- **AWS S3** [https://aws.amazon.com/s3/]

In addition to DynamoDB, Amazon S3 is utilized as the frontend storage solution to store and serve various types of static assets, such as course materials, media files, and user-generated content. S3's integration with other AWS services enables seamless content delivery and efficient data management, optimizing the frontend experience for *EduPath* users.

- **AWS CloudFront [**https://aws.amazon.com/cloudfront/]

Amazon CloudFront acts as a global edge network, caching and delivering static and dynamic content from the nearest edge location to users, reducing latency and improving overall responsiveness. It ensures faster load times, lower bandwidth costs, and improved user experience. This is crucial for our platform as we want a seamless experience for the users as they stream videos, images, and other dynamic content.

- **API-Gateway** [https://aws.amazon.com/api-gateway/]

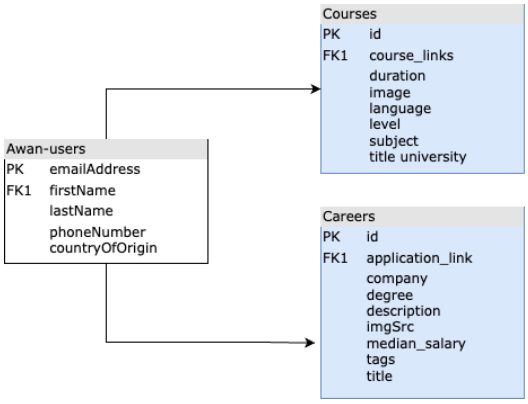
AWS API Gateway is a fully managed service provided by that allows you to create, deploy, and manage APIs. It serves as a front door for your backend services, which in my case AWS Lambda, enabling you to securely expose your APIs to clients and manage their access, traffic, and usage. The API integration enables a smooth and efficient communication channel between web application and the underlying functions, enabling dynamic and interactive user experiences.

Deployment Technologies

- **Vercel [**https://vercel.com/]

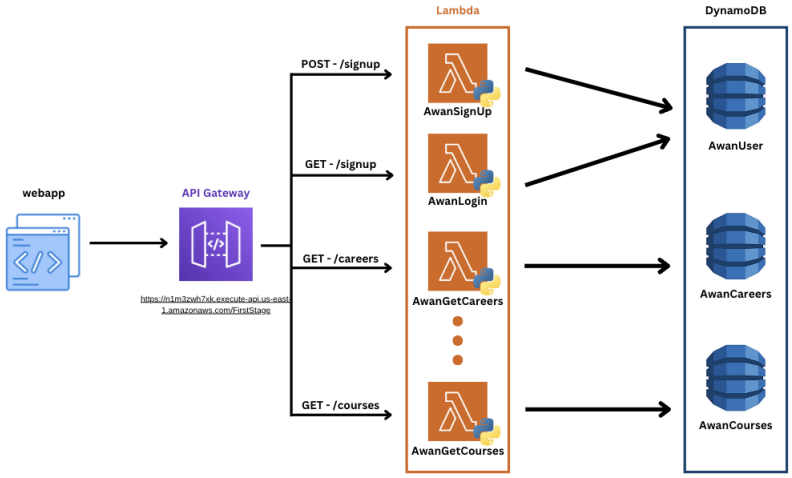
For deployment technology, we decided to use Vercel as it has easy-to-use interface and integration with version control systems like Git, automating the deployment process, ensuring quick and hassle-free updates to the live platform. Vercel's advanced caching and edge network capabilities optimize content delivery, resulting in fast and reliable access to the platform for users worldwide.

Backend



The main purpose of the backend of the project is to manage user authentication, organize educational content, and process and analyze user data for insights. As shown in the schema above, we have three database tables: *awan-users* to manage users, *courses* to manage all the courses displayed in the platform, and the *careers* table to store all the careers available.

The database stands alone with each other, where the *careers* and *courses* tables don't interact. However, for every user, there is a list of courses that they are enrolled in and a list of careers they applied to.



The backend has been fully migrated into Amazon Cloud Service. As can be seen in the diagram above, the backend functions are deployed in the Amazon Lambda, where the code is written in Python 3.10. Each function communicates with a different DynamoDB database to perform different actions. To bridge Amazon Lambda functions to the frontend web application, we use Amazon API Gateway with several endpoints and different request types (POST or GET). The CORS policy for every endpoint has been set to use a wild asterisk in order to perform a successful communication between the Vercel cloud provider with the AWS cloud functions.

In addition to managing user authentication, organizing educational content, and processing user data, we have also implemented Amazon S3 and CloudFront in backend infrastructure to store and deliver course

videos. With Amazon S3, we have created buckets to securely store course videos, taking advantage of its scalability and durability. These videos are uploaded to the S3 buckets, ensuring reliable storage for our educational content. To optimize content delivery and enhance the streaming experience for our users, we have integrated Amazon CloudFront, a content delivery network. CloudFront caches videos at edge locations worldwide, reducing latency and improving the overall video streaming performance.

Future Improvement

The future improvement is driven by the challenges we face, particularly in utilizing deployment technology within AWS cloud infrastructure. To address this, we are incorporating AWS Amplify [https://aws.amazon.com/amplify/], a development framework and toolset that streamlines the creation of web and mobile applications. Amplify offers a wide range of features and services to expedite development, deployment, and scalability. However, due to limitations imposed by AWS Academy Learner Lab account, which prohibits us from creating a new role (AWS Academy Learner Lab – Foundation Services), we have devised an alternative solution by integrating with Vercel Web deployment.

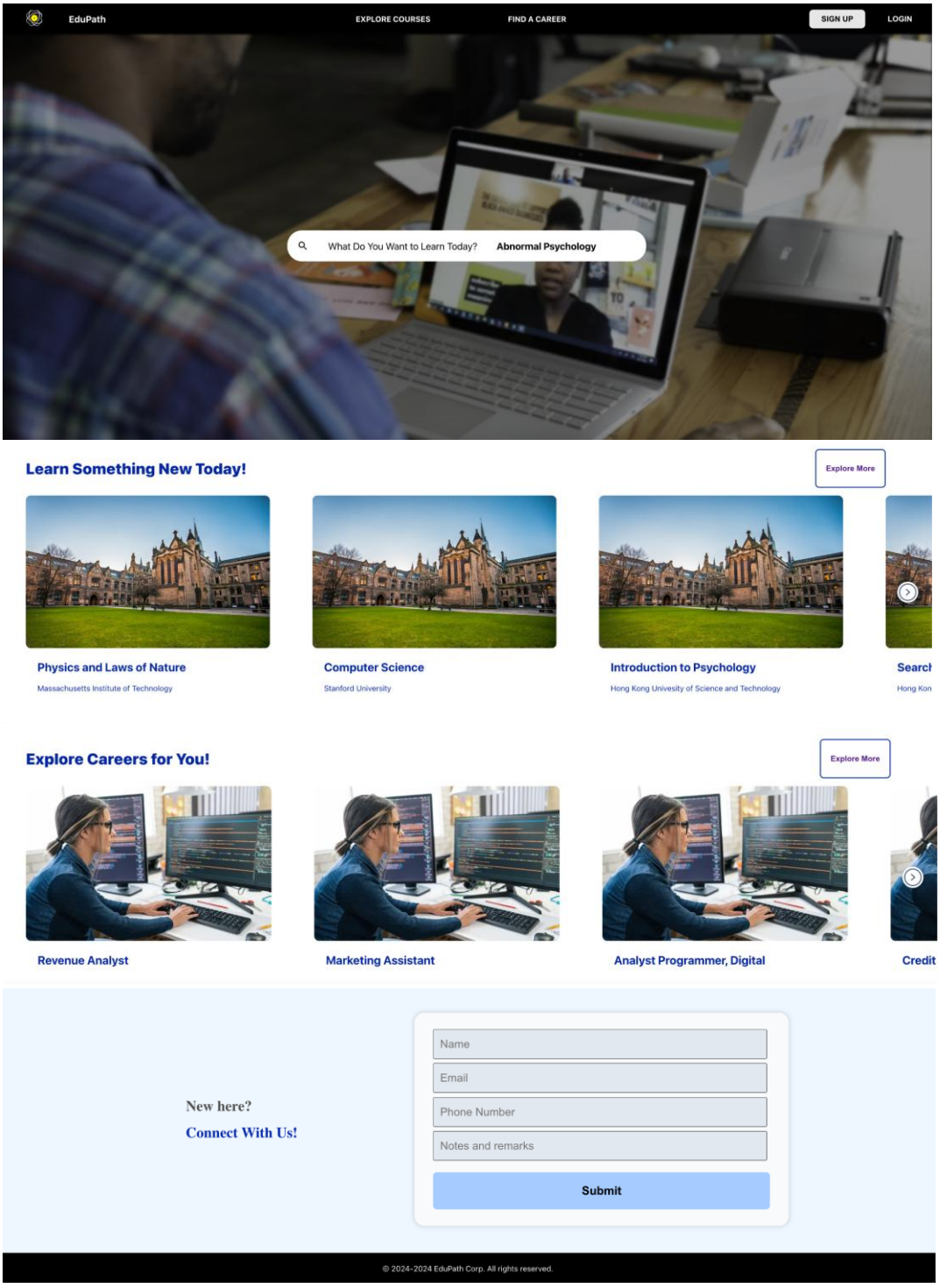
Conclusion

After considering option for the frontend deployment on the cloud services, we have concluded that Vercel is the more suitable choice for the project. Vercel provides a serverless web application deployment which is more suitable. With Vercel's integration with Git, it automates the deployment process if any approved changes occur in the frontend. Instant rollbacks are also enabled through Vercel providing preventive measurements for the platform.

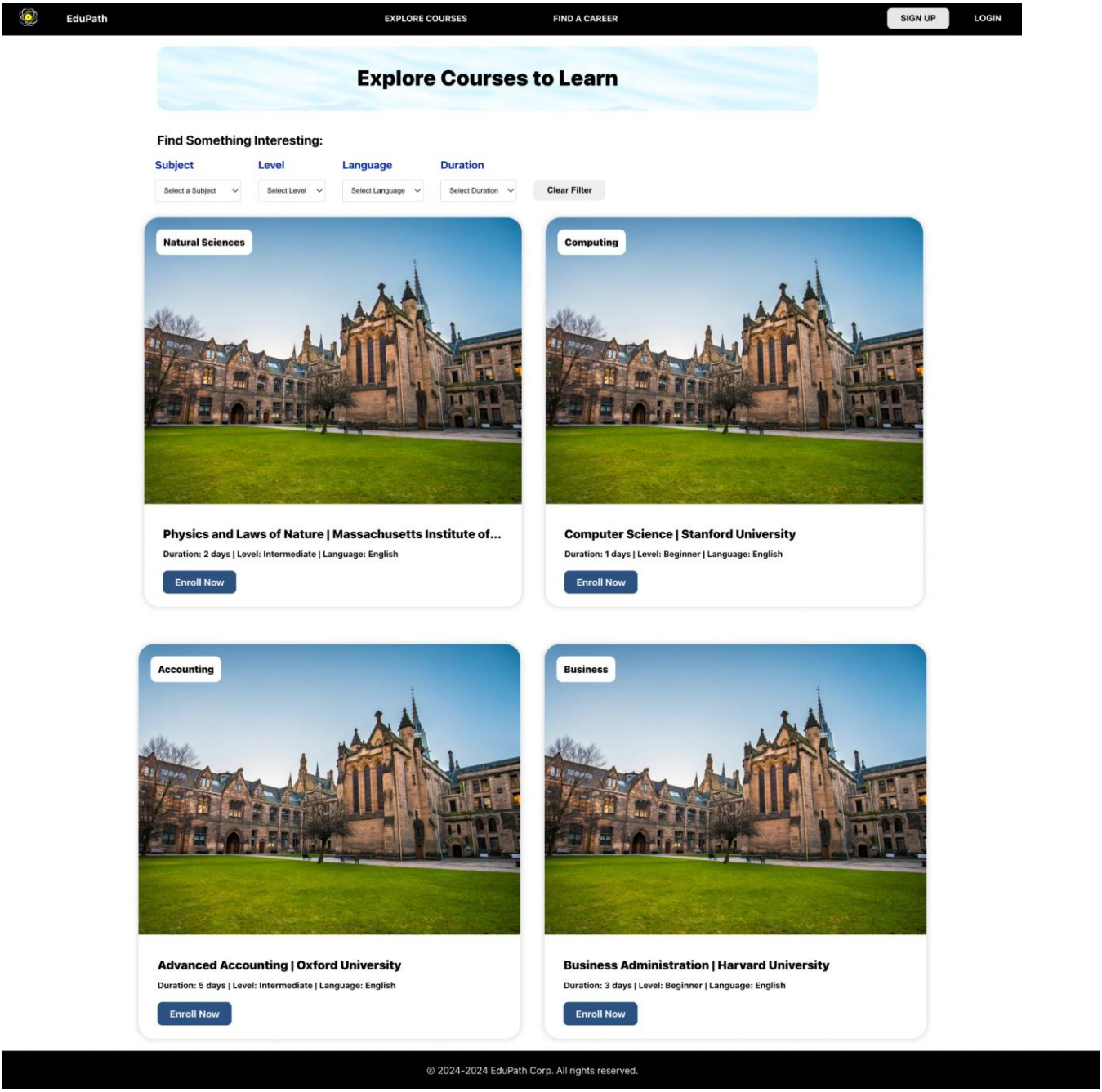
We have chosen Amazon S3 as backend data storage solution due to its cost-effectiveness and efficiency. By integrating Amazon CloudFront with S3, we ensure secure and reliable delivery of files to users. CloudFront's content delivery network capabilities, along with features like HTTPS encryption, access control, and protection against web attacks, provide an additional layer of security for the platform. This combination of S3 and CloudFront creates a robust and secure solution for storing, managing, and delivering content efficiently.

Appendix A [*Our Project Prototype*]

Home Page:



Course Page:



Career Page:

EduPath EXPLORE COURSES FIND A CAREER SIGN UP LOGIN

Find New Careers for You

All Available Careers:

Degree Qualification Job Field Salary

Select Degree | Select Job Field | Select Salary Range | Clear Filter



revenue management | data analysis | inventory management

Revenue Analyst - Cathay Pacific

Median Salary: N/A

Cathay Pacific is seeking an entry-level Revenue Analyst to join our Revenue ..

Apply Now



marketing | campaign management | social media management | project management

Marketing Assistant - AGBA

Median Salary: $35000

AGBA is seeking a Marketing Assistant to join our team in Hong Kong, Hong ...

Apply Now



data engineering | programming languages | database management

Data Engineer - Accenture

Median Salary: N/A

Accenture is seeking a skilled Data Engineer to join our technology team.

Apply Now



data engineering | data modeling | data pipelines | cloud services

Data Engineer - DFI Retail Group

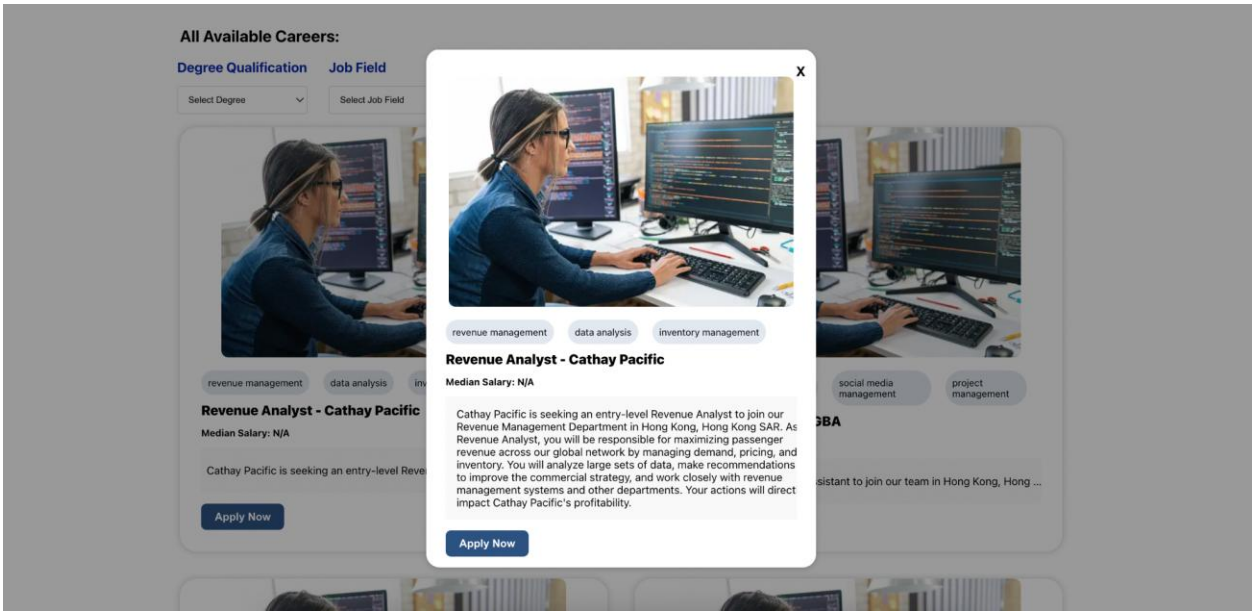
Median Salary: $25000

DFI Retail Group is seeking a Data Engineer to join their team in Quarry Bay, ...

Apply Now

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Modal Career Page:

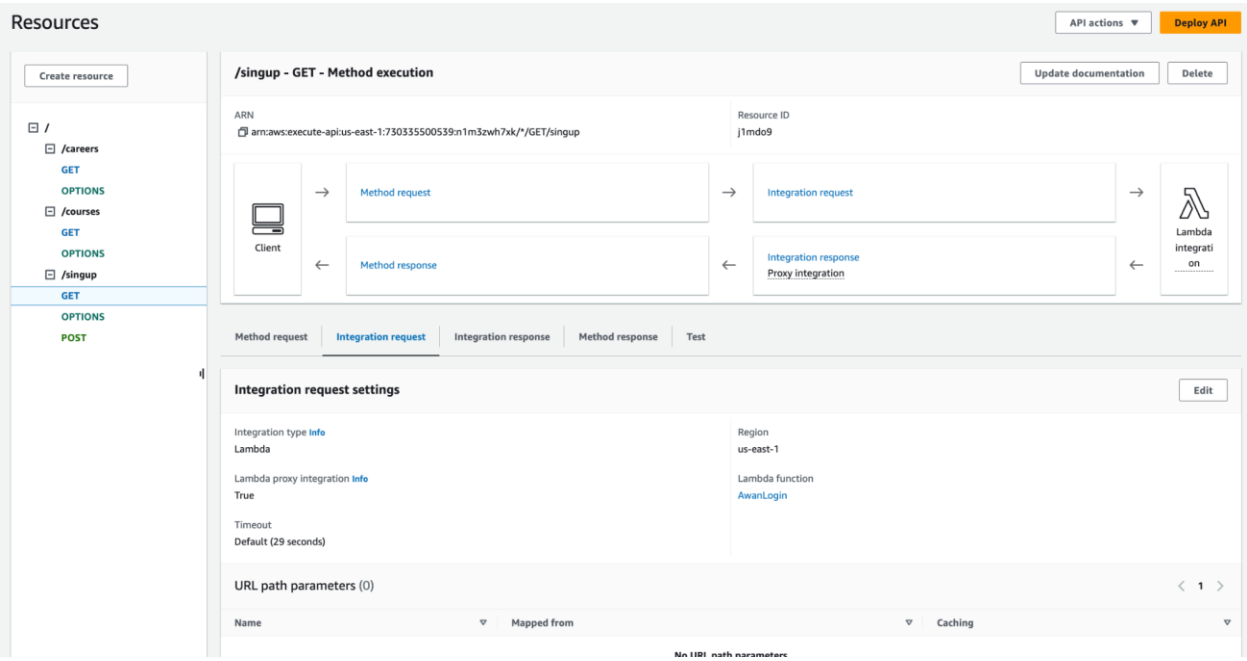


Individual Course Page:

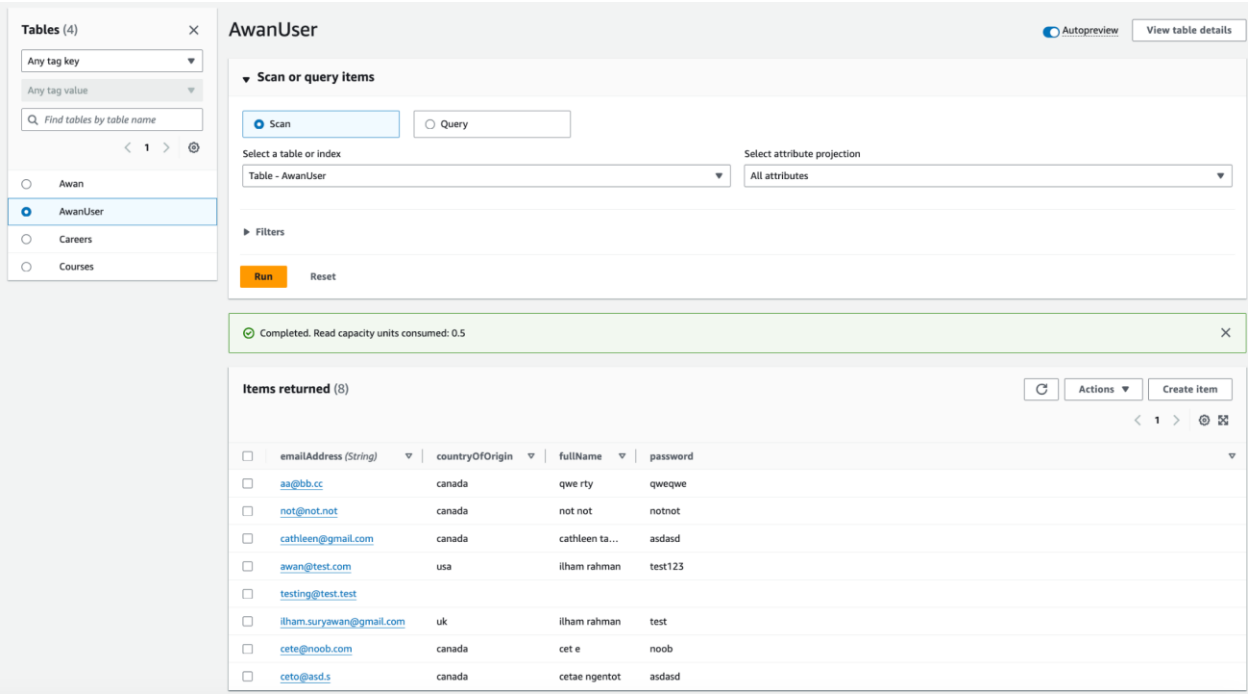


Appendix B [*AWS Implementations*]

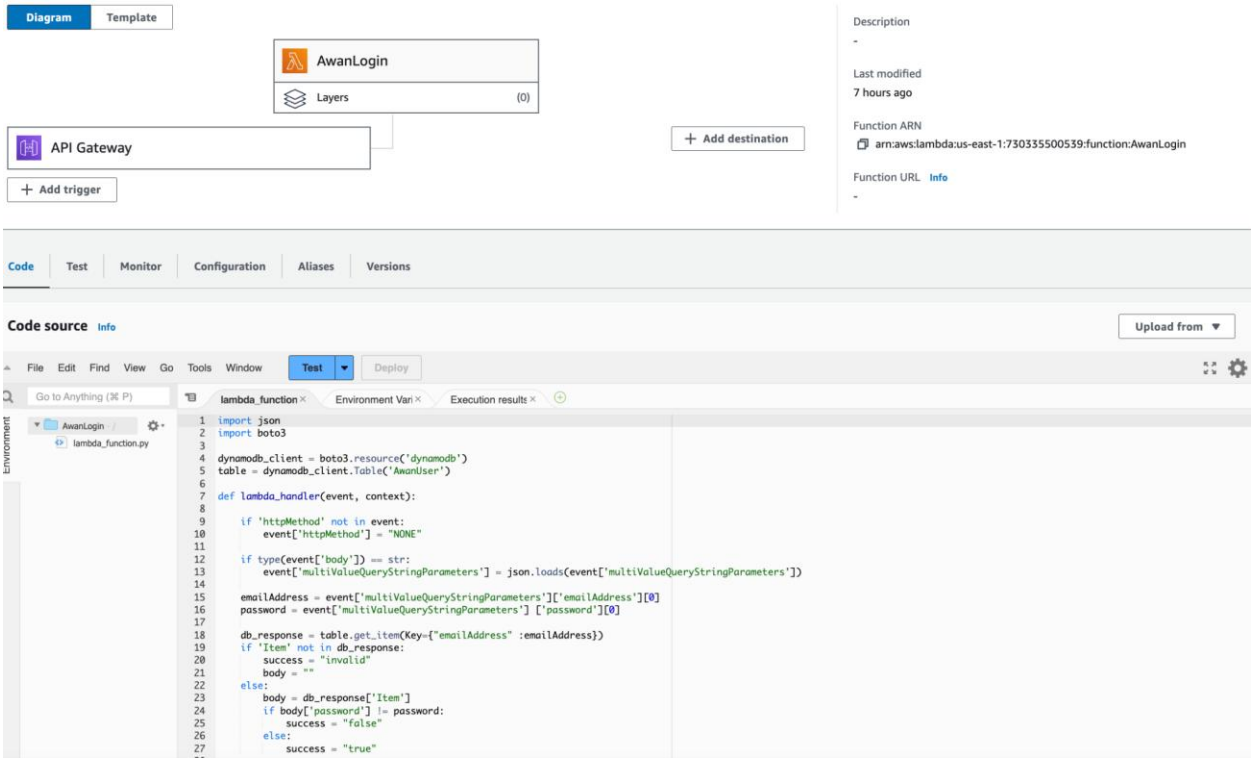
Amazon API Gateaway



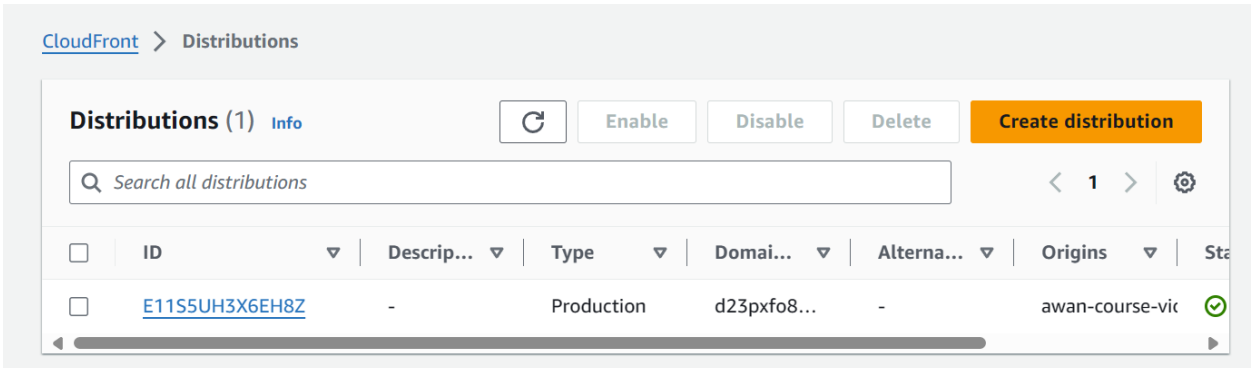
Amazon DynamoDB



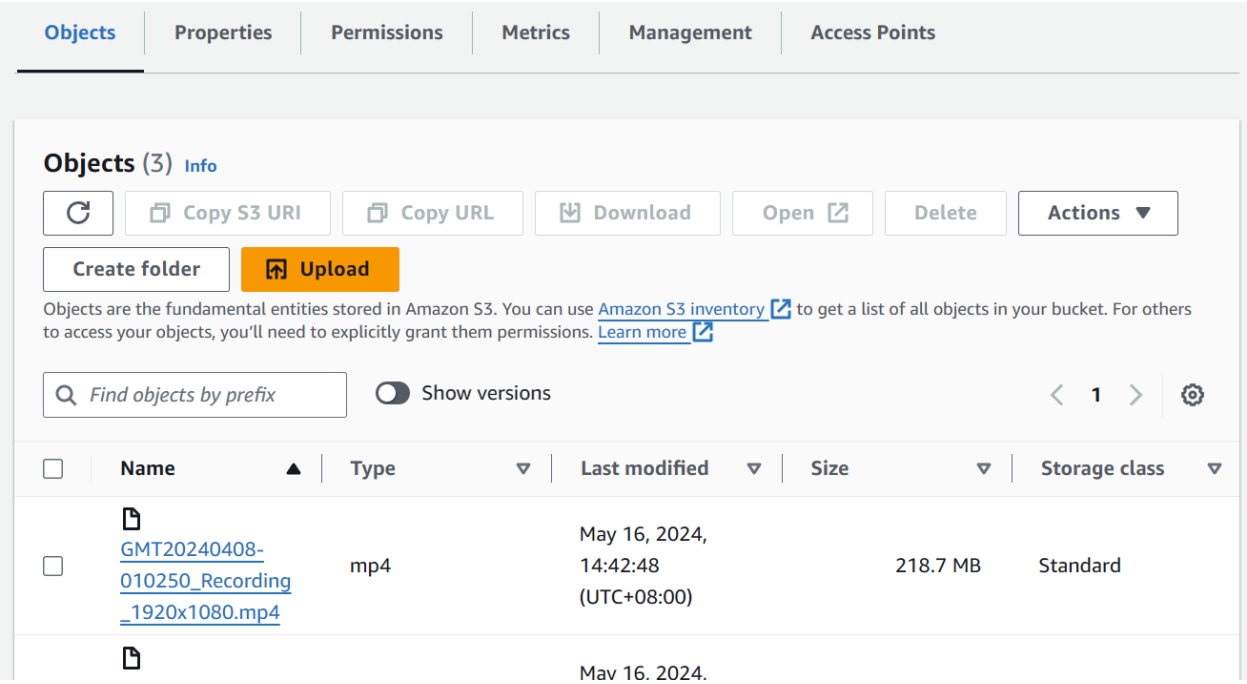
Amazon Lambda



Amazon CloudFront



Amazon S3


Objects
Properties
Permissions
Metrics
Management
Access Points
Objects (3) Info
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Show versions
1
Name
Type
Last modified
Size
Storage class
GMT20240408-010250_Recording_1920x1080.mp4
mp4
May 16, 2024, 14:42:48 (UTC+08:00)
218.7 MB
Standard
May 16, 2024,